

Optimisation et gestion de flux - ST7

H2E Supply Chain Simulation

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April 3, 2026

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Introduction

This report presents a discrete-event simulation model built in Simul8 to dimension the trailer fleet and transport resources of a hydrogen supply chain operating in swap mode. A central Filling Center (FC) equipped with three parallel filling ramps must keep five Hydrogen Refuelling Stations (HRS) continuously supplied, subject to driver shift constraints, limited filling capacity, and variable travel distances. The core question is: *what is the minimum number of trailers that guarantees zero stockouts at every station?*

Section 1 describes the Simul8 model, including the block diagram, the parameters and stochastic distributions used, and the dispatching heuristic that decides when and where to send each trailer. Section 2 addresses warm-up analysis, run length, and model validation. Section 3 presents the sensitivity analysis and experimentation results, and Section 4 answers the four questions posed in the project brief. Implementation details (data structures and Visual Logic code) are documented in Appendix A.

Problem Description

The system follows a cyclical logistics flow: full trailers are dispatched from the Filling Center (FC) to the Hydrogen Refuelling Stations (HRS), where they replace partially consumed trailers via a swap operation. The returned trailers are refilled at the FC, completing the cycle. The challenge is coordinating this flow so that every HRS always has a trailer with sufficient hydrogen, given the constraints on filling capacity, driver availability, and travel times.

Filling Center

The FC operates 24/7 with three parallel filling ramps. A single operator handles all connections and disconnections, attending one ramp at a time; once connected, each trailer fills independently. A trailer that has finished filling remains on its ramp until the operator is available to disconnect it. Each trailer has a maximum capacity of 700 kg, requiring 5 h to fill from empty. The FC includes two parking areas: one for empty trailers awaiting a ramp, and one for full trailers awaiting dispatch.

Hydrogen Refuelling Stations

There are five HRS, each with a fixed, constant consumption rate during a 20-hour daily window (04:00–00:00). Each HRS accommodates exactly one trailer at a time, with no buffer capacity. This assumption will be revisited later. The main parameters are summarised in Table 1.

Parameter	HRS 1	HRS 2	HRS 3	HRS 4	HRS 5
Distance to FC (one-way) [km]	100	150	50	200	75
Hourly consumption [kg/h]	50	50	50	25	10
Daily consumption (20 h/day) [kg/day]	1 000	1 000	1 000	500	200
One-way transit time [h]	2.0	3.0	1.0	4.0	1.5
Round-trip time (no swap) [h]	4.0	6.0	2.0	8.0	3.0

Table 1: Parameters of the five Hydrogen Refuelling Stations.

Transport Fleet and Drivers

Drivers operate in two daily shifts: 05:00–13:00 and 13:00–21:00, with initially 4 drivers per shift. All drivers start and end their shift at the FC. No trips occur between 21:00 and 05:00; only filling operations continue during this window. Each driver performs one round trip at a time, travelling from the FC to a single HRS and back.

Swap Operation

A driver departs the FC with a full trailer and travels to the assigned HRS at 50 km/h. A 30-minute swap takes place on arrival: the partially consumed trailer is disconnected, the full trailer is connected, and the truck returns to the FC hauling the old trailer.

Critical Constraint

No HRS may run out of hydrogen at any time. A full trailer must be dispatched with sufficient lead time so that the trailer in service is always swapped before its content reaches zero.

Initial Conditions

At simulation start, each HRS i has n_i full trailers (each at 700 kg) connected. The FC holds N empty trailers and M full trailers, with none in transit. The total fleet size is therefore $\sum_{i=1}^5 n_i + N + M$. In the base case where each HRS accommodates a single trailer ($n_i = 1$ for all i), this reduces to $5 + N + M$. When multiple tank slots per HRS are considered (Sections 3.1.2 and 3.1.3), the fleet grows accordingly: for example, the multi-tank optimal configuration (2,2,2,2,1 slots) yields a total of $9 + 2 + 3 = 14$ trailers. N and M are the parameters to be optimised.

Questions About the Model

The simulation model is designed to answer four key questions:

- 1. How many empty (N) and full (M) trailers are needed at the FC** to guarantee uninterrupted supply to all five HRS?
- 2. How many parking spots** are needed at the FC for empty and full trailers (i.e. the peak simultaneous occupancy of each parking)?
- 3. How much time does each trailer spend at the FC**, and what is the distribution of time across states (waiting, filling, in transit, at HRS)?
- 4. What is the impact of a Sunday delivery ban** on the number of trailers and buffer capacity required?

The decision variables are: the number of empty trailers N , the number of full trailers M , the number of drivers per shift, and the number of trailer slots at each HRS. All other parameters remain fixed.

KPIs to Measure

To evaluate system performance and guide fleet dimensioning, we define the following KPIs.

1. **Number of stockouts per HRS.** The number of times each HRS reaches a hydrogen level of zero. The primary feasibility indicator: acceptable only if zero for all stations over the entire simulation horizon.
2. **Average and maximum queue size in each parking.** The average reveals whether trailers accumulate excessively at either parking area; the maximum directly determines the number of physical spots required at the FC.
3. **Average trailer residence time at the FC,** broken down into three components: waiting time in the empty parking, time on the filling ramps, and waiting time in the full parking. The sum of the three components gives the total FC cycle time per trailer.
4. **Total completed swap jobs.** The aggregate number of deliveries across all five HRS over the simulation horizon. Used in the sensitivity analysis to quantify the productivity impact of each resource increment.
5. **Utilisation rate of drivers.** The fraction of shift time during which drivers are actively performing trips. Values below 70% indicate over-dimensioning; above 90% indicate fragility.
6. **Utilisation rate of the FC operator.** The fraction of time the operator spends connecting or disconnecting trailers. High utilisation signals that the operator may become a system bottleneck.

Success criteria. A configuration is considered successful if it satisfies all of the following conditions simultaneously:

1. No HRS runs out of hydrogen at any point during the simulation (zero stockouts).
2. The daily hydrogen consumption demand at every station is fully met.
3. The average number of trailers in each parking queue remains well below the total number of trailers in the system.
4. The utilisation rate of both drivers and the operator falls within the 70–90% range. Utilisation below 70% suggests over-dimensioning of resources; utilisation above 90% indicates fragility, where any disruption could cascade into stockouts.

1. Simul8 Model Used

1.1. Block Diagram

Figure 1 shows the complete Simul8 model layout. The left-hand side represents the Filling Center, while the right-hand side represents the five Hydrogen Refuelling Stations and the transport links connecting them to the FC.

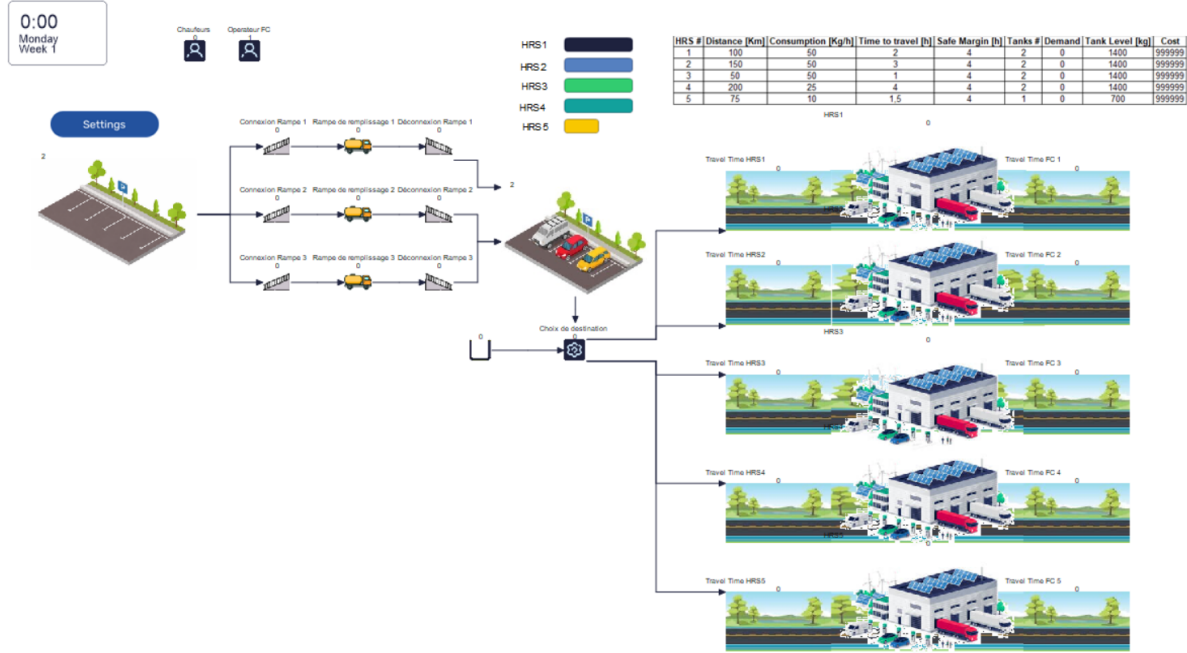


Figure 1: Simul8 model layout.

Trailers entering the FC from the empty trailer parking are routed to one of three parallel filling lines. Each line consists of three sequential stages: a connection activity (Connexion Rampe), a filling activity (Rampe de remplissage), and a disconnection activity (Déconnexion Rampe). The connection and disconnection activities share a single resource, the FC operator, meaning that only one of these six activities can be performed at a time across all three ramps. The filling activity itself requires no operator and runs independently once the trailer is connected.

Once a trailer is fully filled and disconnected, it enters the full trailer parking, where it waits until a driver and a destination are available. The routing decision block (Choix de destination) assigns each trailer to one of the five HRS based on the dispatching heuristic described in Section 1.3.

Each HRS is modelled with two travel activities (one for the outbound trip and one for the return trip) flanking the station itself. A trailer travels from the FC to the assigned HRS, undergoes a swap, and the previously connected trailer is hauled back to the FC on the return trip. Upon arrival at the FC, the returned trailer re-enters the empty trailer parking, closing the cycle.

Two resources govern the system: the FC operator, who is available 24/7 and handles all connections and disconnections at the ramps, and the pool of drivers (Chauffeurs), whose

availability is limited by shift schedules. A settings table in the model interface allows the user to modify key parameters for each HRS, including the number of tanks, the safe margin, and the initial tank level.

1.2. Parameters and Stochastic Distributions

Table 2 summarises the key parameters and distributions used in the Simul8 model. All time-based quantities are expressed in minutes, which is the internal time unit of the simulation.

Parameter	Distribution / Value	Unit
Trailer capacity	700 (fixed)	kg
Filling time (general)	$300 \cdot (700 - r)/700$	min
Operator connect/disconnect time	Uniform(3, 8)	min
Swap time at HRS	30 (fixed)	min
Travel time to/from HRS 1	Normal(120, 12)	min
Travel time to/from HRS 2	Normal(180, 18)	min
Travel time to/from HRS 3	Normal(60, 6)	min
Travel time to/from HRS 4	Normal(240, 24)	min
Travel time to/from HRS 5	Normal(90, 9)	min
HRS operating window	04:00 – 00:00	–
Day shift	05:00 – 13:00	–
Night shift	13:00 – 21:00	–
Drivers per shift	4 (initial assumption)	–

Table 2: Parameters and distributions used in the Simul8 model.

Since trailers could come back to the FC hauling half-empty tanks, we have made the assumption that it affects the refuelling time in the FC. A completely empty trailer takes exactly 300 minutes (5 hours) to fill, while a trailer with residual content takes proportionally less:

$$t_{\text{fill}} = 5 \cdot \frac{700 - r}{700} \quad [\text{hours}] \quad (1)$$

The only stochastic elements in the model are the operator’s connection and disconnection time and the travel times. In the absence of data or indications about the probabilistic distributions that these operations follow, we chose to model travel times using a Normal distribution with a standard deviation equal to 10% of the mean, which we considered a reasonable representation of real-world variability in road transport. The time that it takes for the operator to connect and disconnect follows a uniform distribution between 3 and 8 minutes.

Both distributional choices were discussed and validated during the intermediate project review meeting.

1.3. Dispatching Heuristic

The dispatching logic determines *when* to send a trailer and *to which* HRS. Rather than scheduling trips in advance, the model uses a reactive, greedy heuristic: at each evaluation cycle during HRS operating hours, it computes a delivery cost for every candidate station and dispatches a trailer to the one with the lowest cost.

At each cycle, the algorithm performs two operations in sequence. First, it decrements the hydrogen level at every HRS by the station’s hourly consumption unconditionally, regardless of whether a delivery is already en route to that station. Second, for each station whose demand flag is zero (meaning no delivery is currently in progress), it evaluates a cost function.

Cost function. For each eligible HRS i , a cost C_i is computed as:

$$C_i = C_{\text{travel},i} + C_{\text{ineff},i} + C_{\text{overtime},i} + C_{\text{rupture},i} \quad (2)$$

The first three terms are non-negative penalties. The fourth term, the rupture component, can be either strongly negative (for stations in danger of running dry) or positive (for stations with comfortable reserves). The station with the lowest resulting cost C_i is selected as the dispatch target.

- **Travel cost** ($C_{\text{travel},i}$): the baseline one-way travel cost for a delivery to HRS i , pre-computed at initialisation as $t_i \times c_{\text{travel}}$, where t_i is the one-way travel time and $c_{\text{travel}} = 50$ /h is the driver’s hourly wage.
- **Inefficiency penalty** ($C_{\text{ineff},i}$): penalises deliveries where the station cannot absorb a full 700 kg trailer because its tank level is still too high. The algorithm projects the tank level at the moment the trailer would arrive: $L_i^{\text{arr}} = L_i - t_i \cdot q_i$, where L_i is the current level, t_i the one-way travel time, and q_i the consumption rate. The free capacity at arrival is $f_i = n_i \cdot 700 - L_i^{\text{arr}}$ (where n_i is the number of trailer slots at the station), and the effectively delivered quantity is $d_i = \min(f_i, 700)$. The penalty is:

$$C_{\text{ineff},i} = (700 - d_i) \times c_{\text{ineff}}$$

The undelivered quantity $(700 - d_i)$ kg must be vented or returned unconsumed, representing a direct loss at the market price of hydrogen. Based on current wholesale hydrogen prices, we set $c_{\text{ineff}} = 2$ /kg. If the full 700 kg can be absorbed, this term is zero.

- **Overtime cost** ($C_{\text{overtime},i}$): penalises trips whose projected return time exceeds the end of the driver’s current shift. The projected return hour is $h_{\text{return}} = h_{\text{now}} + 2t_i + 0.5$ (round trip plus 30-minute swap). The shift boundary $h_{\text{shift_end}}$ is 21:00 during the night shift and 13:00 during the day shift. The overtime cost is:

$$C_{\text{overtime},i} = \max(0, h_{\text{return}} - h_{\text{shift_end}}) \times c_{\text{overtime}}$$

The overtime rate is set to $c_{\text{overtime}} = 75$ /h, reflecting the standard overtime premium applied to the base driver wage of $c_{\text{travel}} = 50$ /h. If the driver returns before the shift ends, this term is zero.

- **Rupture component** ($C_{\text{rupture},i}$): this is the key mechanism that forces the algorithm to prioritise stations at risk of running dry while discouraging unnecessary early deliveries. The projected arrival level L_i^{arr} is compared against the safe margin expressed in kilograms ($\sigma_i \cdot q_i$, since σ_i is in hours and q_i in kg/h). The autonomy of the station at arrival, expressed in hours, is $a_i = L_i^{\text{arr}}/q_i$:

$$C_{\text{rupture},i} = \begin{cases} -B & \text{if } L_i^{\text{arr}} \leq \sigma_i \cdot q_i \\ (a_i - \sigma_i) \times c_{\text{surplus}} & \text{otherwise} \end{cases}$$

In the urgent case, the large bonus B is subtracted from the total cost, driving C_i strongly negative and guaranteeing that this station is selected. In the non-urgent case, the surplus autonomy beyond the safe margin (in hours) is penalised at rate $c_{\text{surplus}} = 2$ /h: a trailer dispatched too early will arrive at a station that still has ample reserve, tying up a trailer and a driver unproductively. The penalty therefore discourages premature deliveries and encourages the heuristic to wait until the station can absorb a larger share of the incoming 700 kg load.

Safe margin. The safe margin σ_i (expressed in hours) determines the minimum hydrogen reserve that should remain at HRS i when a delivery arrives. For comparison with the tank level (in kg), it is converted via $\sigma_i \cdot q_i$. The margin is computed dynamically depending on the current shift and day of the week:

- **Night shift, Monday to Friday:** $\sigma_i = 3t_i + 4$. The multiplier of 3 on the one-way travel time provides a generous safety buffer, and the additive floor of 4 hours ensures a minimum reserve even for nearby stations. This formula is used during the night shift because it is the last driving window before the overnight gap (21:00–05:00): the margin must be large enough to guarantee that no station runs dry before the first dispatch of the following morning.
- **Night shift, Saturday:** $\sigma_i = 3t_i + 4 + 24$. An extra 24 hours are added to pre-load the stations with enough hydrogen to survive the entirety of Sunday, during which the dominical restriction prohibits all deliveries.
- **Day shift and outside driver shifts:** $\sigma_i = \sigma_{\text{default}}$, a fixed global parameter. During the day shift, the upcoming night shift can still make additional deliveries, so a less aggressive margin suffices. Outside driver shifts (21:00–05:00), the margin is recorded for monitoring but has no operational effect since no drivers are available to dispatch.

Dispatch conditions. After evaluating all five stations, the one with the minimum C_i is selected. A trailer is dispatched only if all of the following conditions hold:

1. The minimum cost is below a threshold of 5 000, which prevents unnecessary trips when no station truly needs hydrogen (this threshold sits above the maximum possible travel cost but far below the rupture bonus B , so urgent deliveries always pass).
2. At least one full trailer is available in the full parking.
3. At least one driver is currently free.

If any condition fails, no dispatch occurs and the system waits for the next evaluation cycle.

A **demand flag** per HRS prevents duplicate dispatches: once a trailer is sent to a station, its flag is set to 1, excluding it from cost evaluations until the trailer completes the round trip and returns to the FC. Only one trailer can be dispatched per evaluation cycle; if multiple stations are simultaneously critical, they are addressed in successive cycles in order of ascending cost.

Numerical Example: HRS 1 at 14:00 (Night Shift)

Parameters: $t_1 = 2$ h, $q_1 = 50$ kg/h, $\sigma_1 = 3 \times 2 + 4 = 10$ h, $L_1 = 400$ kg, $C_{\text{travel},1} = 2 \times 50 = 100$, $c_{\text{ineff}} = 2$ /kg, $B = 100\,000$, $c_{\text{surplus}} = 2$ /h.

Projected arrival level:	$L_1^{\text{arr}} = 400 - 2 \times 50 = 300$ kg
Safe margin in kg:	$\sigma_1 \cdot q_1 = 10 \times 50 = 500$ kg
Urgency check:	$300 \leq 500 \Rightarrow \text{urgent}, C_{\text{rupture}} = -100\,000$
Free capacity:	$f_1 = 700 - 300 = 400$ kg, $d_1 = 400$ kg
Inefficiency:	$(700 - 400) \times 2 = 600$
Overtime:	$h_{\text{return}} = 14 + 4.5 = 18.5 < 21 \Rightarrow 0$
Total cost:	$100 + 600 + 0 + (-100\,000) = -99\,300$

The strongly negative cost guarantees that HRS 1 will be selected over any non-urgent station. For a non-urgent station with $L^{\text{arr}} = 800$ kg, $q = 50$ kg/h, and $\sigma = 10$ h, the autonomy would be $a = 800/50 = 16$ h, yielding $C_{\text{rupture}} = (16 - 10) \times 2 = 12$, a positive penalty that discourages an early delivery.

2. Obtaining Reliable Results

2.1. Setting up the Warm-Up Period

Before using the model to answer the project questions, we estimated the warm-up period using the graphical method. The hydrogen level at each HRS was selected as the variable to analyse, and its evolution over time was plotted. The goal is to observe how long the system takes to reach a stable, cyclical behaviour. A single, long run was performed so that the fluctuations in the indicator could be clearly appreciated and an appropriate warm-up period could be identified.

To reduce the volatility of the raw data and make it easier to spot the transition to steady state, we applied a moving-average filter with a window of $k = 10$ observations to the hydrogen-level time series obtained from the long run. Results are presented for the optimal configurations obtained later in Section 3, both for the scenario in which Sunday deliveries are allowed and for the scenario in which they are banned. The simulation outputs are shown in Figure 2.

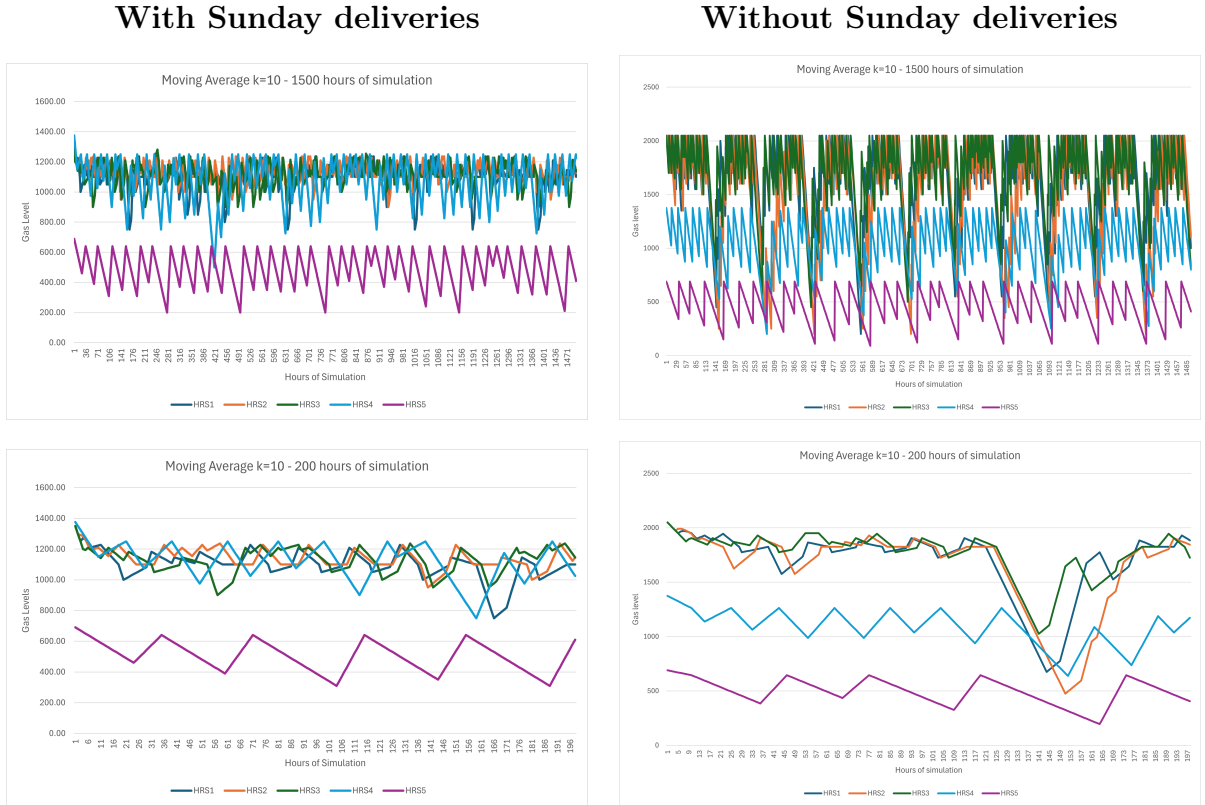


Figure 2: Moving average ($k = 10$) of hydrogen levels at each HRS. Top row: 1 500 hours of simulation. Bottom row: close-up of the first 200 hours.

The plots reveal that hydrogen levels follow a cyclical pattern that settles into a stationary regime very quickly from the start of the simulation. In other words, the warm-up period is very short compared to the total run length.

More specifically, the close-up views (bottom row) suggest that the warm-up period falls between 40 and 60 hours. It should be noted, however, that this estimate is inherently

subjective: even after applying the moving average, the data remain quite volatile, and the gas levels converge rapidly to a steady state without a clearly identifiable transition point.

To be on the safe side, a warm-up period of **60 hours** was adopted and entered into the Simul8 *Warm-Up Period* field in the simulation settings. This ensures that all statistics collected by the model (queue sizes, utilisation rates, residence times, and stockout counts) are computed exclusively from the steady-state portion of each replication, with the initial transient automatically discarded by Simul8 before any results are recorded.

2.2. Number of Runs

Good practice in simulation theory requires running several replications of the same experiment to avoid biased results and to obtain statistically meaningful estimates. Using the Simul8 Trial Calculator with a 95% confidence interval, the recommended number of replications was 4. This is adopted as the standard practice for all experiments and tests presented in this report.

2.3. Validation of the Model

All deterministic assumptions (consumption profiles, shift schedules, trailer capacity, swap duration) are those transmitted in the original problem statement. The only stochastic elements are the operator’s connection/disconnection time, and the travel times. These choices were made to capture realistic variability in road transport and manual operations without introducing too much complexity. In the absence of empirical data, both distributions were discussed and validated during the intermediate project review meeting.

White-box validation. Given the complexity and difficulty of use of Visual Logic, we performed a series of internal verification tests to confirm that the Simul8 implementation faithfully reproduces the intended logic:

1. **Manual trailer tracking.** The model was run in step-by-step mode, following a several trailers through their complete cycle to observe if all HRS were covered and the return of the trailers was done correctly.
2. **Hydrogen level verification.** In step-by-step mode, we checked that all the HRS gas levels decreased each hour and that if they arrived at 0, they would remain at zero (avoiding negative values of gas).
3. **Extreme conditions.** With $N = 0$ and $M = 0$ (no trailers at the FC), the model rapidly triggered the stockout alert “HRS à sec” at every station. Conversely, with $N = 50$ and $M = 50$, no stockout was ever observed. Both outcomes confirm that the rupture detection mechanism and the dispatching heuristic behave as intended under boundary conditions.
4. **Mass balance.** At several random instants during the simulation, the total number of trailers across all states (connected at HRS, in transit in both directions, in Parking VIDE, on a filling ramp, and in Parking PLEIN) was counted and verified to equal the constant $5 + N + M$, confirming that no trailers are created or destroyed during the simulation.

Black-box validation. Since there is no real system to observe, no empirical output data are available, and a direct statistical comparison is therefore not feasible. The validity of the model rests on the correctness of the conceptual model, the distributional choices discussed above, and the white-box verification tests.

3. Experimenting

The previous sections established the model structure, the dispatching heuristic, and the validation methodology. This section uses the model to identify the minimum resource configuration that guarantees zero stockouts, and then explores how the system responds to variations in driver and trailer availability.

3.1. Finding Optimal Parameters

The starting point for the experimentation is the configuration given by the original problem statement: 4 drivers per shift, a single tank at each HRS, and a number of starting trailers at the FC equal to the number of drivers per shift (i.e. 4 empty and 4 full). Sunday deliveries are allowed.

Running this configuration for 5 years of simulated time confirmed that the dispatching heuristic maintains uninterrupted supply at all five stations. Since the system is feasible with this generous endowment, we proceed by sequentially reducing the available resources, first trailers and then drivers, until the system can no longer guarantee supply. At each step, only the parameter shown in bold is changed relative to the previous row.

3.1.1 Experiments with Standard Buffer Capacity at the HRS

In this first round of experiments, every HRS holds exactly one tank. Table 3 shows the configurations tested and whether each one maintained uninterrupted supply over the 5-year horizon.

Works?	Starting Trailers		Drivers per Shift		Tanks per HRS				
	Empty	Full	Day	Night	HRS1	HRS2	HRS3	HRS4	HRS5
✓	4	4	4	4	1	1	1	1	1
✓	3	4	4	4	1	1	1	1	1
✓	3	3	4	4	1	1	1	1	1
✓	3	3	3	4	1	1	1	1	1
×	3	3	3	3	1	1	1	1	1
×	3	3	4	3	1	1	1	1	1
✓	2	3	3	4	1	1	1	1	1
×	2	2	3	4	1	1	1	1	1
✓	3	2	3	4	1	1	1	1	1

Table 3: Sequential resource reduction with a single tank per HRS (Sunday deliveries allowed).

The optimal configuration under this constraint is shown in Table 4.

Starting Trailers		Drivers per Shift		Tanks per HRS				
Empty	Full	Day	Night	HRS1	HRS2	HRS3	HRS4	HRS5
2	3	3	4	1	1	1	1	1

Table 4: Optimal configuration with single-tank HRS and Sunday deliveries: (2 empty, 3 full / 3 day, 4 night / 1,1,1,1,1 tanks).

Two key findings emerge. First, the system requires at least 5 trailers at the FC; the split between parkings is interchangeable, since both (2 empty, 3 full) and (3 empty, 2 full) work, but dropping to 4 total always causes stockouts. Second, the night shift must have strictly more drivers than the day shift. This is because the night shift is the last dispatch window before the overnight gap (21:00–05:00) and must pre-position enough hydrogen to cover consumption until the first morning delivery.

3.1.2 Experiments Increasing Buffer Capacity at the HRS

Allowing each HRS to hold more than one tank simultaneously provides additional on-site buffer. Starting from the previous optimal and setting all HRS to 2 tanks, we again reduce resources sequentially (Table 5).

Works?	Starting Trailers		Drivers per Shift		Tanks per HRS				
	Empty	Full	Day	Night	HRS1	HRS2	HRS3	HRS4	HRS5
✓	2	3	3	4	2	2	2	2	2
✓	2	3	3	3	2	2	2	2	2
✓	2	3	2	3	2	2	2	2	2
×	2	2	2	3	2	2	2	2	2
				⋮					
✓	2	3	2	3	2	2	2	2	1

Table 5: Resource reduction with multiple tanks per HRS (Sunday deliveries allowed).

The additional buffer allows the system to operate with fewer drivers. The optimal configuration is shown in Table 6.

Starting Trailers		Drivers per Shift		Tanks per HRS				
Empty	Full	Day	Night	HRS1	HRS2	HRS3	HRS4	HRS5
2	3	2	3	2	2	2	2	1

Table 6: Optimal configuration with multi-tank HRS and Sunday deliveries: (2 empty, 3 full / 2 day, 3 night / 2,2,2,2,1 tanks).

HRS 5 does not require a second tank: its low consumption rate (10 kg/h) gives each trailer over 70 hours of autonomy, far exceeding the overnight gap.

3.1.3 Experimenting with Sunday Restrictions

When Sunday deliveries are prohibited, the system must accumulate enough hydrogen during Saturday to cover Sunday’s full 20-hour consumption window without replenishment. Running the simulation under the Sunday ban with the original baseline (4 empty, 4 full / 4 day, 4 night / 1,1,1,1,1 tanks) results in stockouts. Both optimal configurations from Sections 3.1.1 and 3.1.2 also fail.

The natural solution is to increase the number of tanks at the HRS to build a larger on-site buffer. Following a procedure analogous to the ones described above, and targeting the same trailer and driver counts as the multi-tank optimal (2 empty, 3 full / 2 day, 3 night drivers), the minimum tank configuration that avoids stockouts under the Sunday ban is shown in Table 7.

Starting Trailers		Drivers per Shift		Tanks per HRS				
Empty	Full	Day	Night	HRS1	HRS2	HRS3	HRS4	HRS5
2	3	2	3	3	3	3	2	1

Table 7: Optimal configuration under the Sunday delivery ban: (2 empty, 3 full / 2 day, 3 night / 3,3,3,2,1 tanks).

The three high-consumption stations (HRS 1 to 3, 1 000 kg/day each) require 3 tanks to buffer the Sunday gap. HRS 4 (500 kg/day) needs 2, while HRS 5 (200 kg/day) remains at 1.

3.2. Sensitivity Analysis

To understand how the system responds to changes in driver and trailer availability beyond the optimal configurations, we conducted a sensitivity analysis varying these two resources across five scenarios. All scenarios use the multi-tank configuration (2 empty, 3 full / variable drivers / 2,2,2,2,1 tanks) with Sunday deliveries allowed.

3.2.1 Scenarios Evaluated

Scenario	Description	Drivers (Day / Night)	Trailers
A	Baseline	5 (2 / 3)	5
B	Extra trailer	5 (2 / 3)	6
C	Extra driver (v1)	6 (3 / 3)	5
D	Extra driver (v2)	6 (2 / 4)	5
E	Both increased	6 (2 / 4)	6

Table 8: Scenarios for the sensitivity analysis. The driver split between the day shift (05:00–13:00) and the night shift (13:00–21:00) is shown in parentheses. Scenarios C and D both have six drivers in total but differ in how they are distributed across shifts: C allocates them symmetrically (3 day / 3 night), while D concentrates the extra driver on the night shift (2 day / 4 night), the last dispatch window before the overnight gap. This distributional difference explains the small but consistent productivity gap between the two scenarios (12 584 vs. 12 687 completed jobs).

Total system productivity is measured as the sum of completed swap jobs across all five HRS. Queue behaviour is captured by the average queue size in the empty and full parkings.

3.2.2 Results

Scenario	Drivers	trailers	Total Jobs
A	5	5	11 982
B	5	6	12 223
C	6	5	12 584
D	6	5	12 687
E	6	6	13 792

Table 9: Total completed jobs per scenario.

Scenario	Drivers	trailers	Avg. Queue (Empty)	Avg. Queue (Full)
A	5	5	0.014	2.521
B	5	6	0.016	3.497
C	6	5	0.042	2.437
D	6	5	0.041	2.426
E	6	6	0.054	3.313

Table 10: Average parking queue sizes per scenario.

3.2.3 Key Findings

Adding a trailer without extra drivers (A vs. B). Productivity increased from 11 982 to 12 223 (+2.01%), while the average full-parking queue grew from 2.521 to 3.497 (+38.68%). When driver availability is limited, the additional trailer cannot be dispatched frequently enough and accumulates in the full parking. The system cannot take advantage of the extra vehicle due to the lack of personnel, resulting in a modest throughput gain at the cost of significant congestion.

Adding a driver without extra trailers (A vs. C/D). Productivity rose to 12 584 in Scenario C (+5.02%) and to 12 687 in Scenario D (+5.88%), while the full-parking queue *decreased* slightly to 2.437 and 2.426 respectively. The gain from adding a driver is clearly larger than from adding a trailer. In addition to improving productivity, it also reduces congestion. This confirms that driver availability is the binding constraint at this stage.

Adding a driver with maximum fleet (B vs. E). Productivity increased from 12 223 to 13 792 (+12.84%), while the full-parking queue decreased from 3.497 to 3.313 (−5.25%). This is the largest improvement observed in the entire analysis. It confirms that when trailer capacity is already high, the shortage of drivers is the dominant limiting factor. Relieving this bottleneck improves throughput and reduces congestion simultaneously.

Adding a trailer with maximum drivers (C/D vs. E). Productivity increased by +9.60% relative to Scenario C and by +8.71% relative to Scenario D. This is a critical observation: once the driver constraint is relieved (having 6 drivers), adding a trailer becomes much more effective (8.7%–9.6% improvement) than when only 5 drivers are available (2.01% improvement). This reveals a sequential bottleneck dynamic in the system:

1. Initially, the system is limited by drivers.
2. Once that constraint is eased, trailer capacity becomes the relevant bottleneck.
3. The best results are achieved when both resources increase together.

Flow efficiency caveat. Although Scenario E is the most productive, its full-parking queue (3.313) is larger than in the 6-driver / 5-trailer scenarios (≈ 2.43). Therefore, the best scenario in terms of productivity is not necessarily the most fluid. This trade-off should be considered when selecting the final operational configuration.

3.2.4 Station-Level Analysis: HRS 5

The increase in resources did not affect all stations equally. The most pronounced growth occurred at HRS 5:

Scenario	HRS 5 Jobs
A	2 284
B	2 475
C	2 764
D	2 816
E	3 609

Table 11: Jobs completed at HRS 5 per scenario.

The percentage impacts at HRS 5 are notable. Going from 5 to 6 drivers while keeping 5 trailers increased HRS 5’s output by 21.02% (Scenario C) and 23.29% (Scenario D). Going from 5 to 6 drivers with 6 trailers yielded a 45.82% increase. Adding a sixth trailer with 6 drivers already in place improved output by 30.57% (vs. C) and 28.16% (vs. D).

HRS 5 is particularly sensitive to the availability of transport resources and benefits enormously from additional dispatch flexibility. Its low consumption rate (10 kg/h) causes the dispatching heuristic to systematically deprioritise it under tight resource conditions, since the cost function favours stations closer to running dry. When resources become more abundant, HRS 5 is the first station to capture the freed-up capacity. This behaviour deserves further investigation in future refinements of the model.

3.2.5 Economic Considerations

The best-performing scenario from an operational standpoint is not automatically the best managerial decision. Each increment in resources generates additional costs: driver salaries, social charges, training, fuel, vehicle maintenance, and insurance.

Scenario E (6 drivers, 6 trailers) offers the highest productivity, but it also demands the largest capital and operational investment. Scenarios C and D (6 drivers, 5 trailers) may offer a better cost-benefit balance if the marginal productivity gained from the sixth trailer does not justify its associated expenditure. The productivity gap between C/D and E (approximately 8–10%) must be weighed against the full cost of acquiring and operating an additional vehicle.

The final decision should therefore integrate the operational results presented in this simulation with the financial viability and profitability of each scenario.

4. Answers to Basic Questions

4.1. Number of Empty and Full Trailers at the Start

Through the sequential reduction procedure described in Section 3, we identified two viable configurations when Sunday deliveries are allowed: one with a single tank slot per HRS, and one allowing up to two tanks at each station (except HRS 5). Table 12 compares both.

Configuration	Trailers at FC		Drivers / shift		Tanks per HRS			Total trailers in system
	Empty	Full	Day	Night	HRS 1–3	HRS 4	HRS 5	
Single-tank HRS	2	3	3	4	1	1	1	10
Multi-tank HRS	2	3	2	3	2	2	1	14

Table 12: Optimal fleet configurations with Sunday deliveries allowed.

Both configurations require exactly 5 trailers at the FC (2 empty, 3 full); reducing to 4 always causes stockouts regardless of distribution. The night shift requires more drivers than the day shift because it is the last dispatch window before the overnight gap (21:00–05:00) and must pre-position enough hydrogen to cover consumption until the first morning delivery reaches the most distant HRS.

Recommended configuration We recommend the **multi-tank configuration** (2 empty, 3 full / 2 day, 3 night / 2,2,2,2,1 tanks) as the optimal solution, for three reasons.

First, it is the result of an exhaustive elimination procedure: every neighbouring configuration with fewer resources produces stockouts, certifying this as the minimum feasible combination across all dimensions tested.

Second, the multi-tank configuration reduces the driver requirement from 7 to 5 drivers per day compared to the single-tank alternative, at the cost of 4 additional trailers parked at the HRS. Since drivers represent a recurring operational expense while parked trailers are a one-time capital investment, this trade-off is economically favourable. The additional on-site buffer also absorbs travel-time variability, transforming a fragile system into a robust one.

Third, the total fleet of 14 trailers is lean for a system with 3 700 kg of aggregate daily demand (equivalent to 5.3 full trailers per day), confirming that the dispatching heuristic efficiently utilises the fleet without immobilising capital unnecessarily.

4.2. Number of Parking Spots Needed

Table 13 shows the parking queue statistics for the optimal single-tank configuration, obtained over 4 replications at 95% confidence.

Parking	Metric	Low 95%	Average	High 95%
Parking VIDE	Average Queue Size	0.00	0.00	0.01
Parking VIDE	Maximum Queue Size	2.00	2.00	2.00
Parking PLEIN	Average Queue Size	2.31	2.31	2.31
Parking PLEIN	Maximum Queue Size	5.00	5.00	5.00

Table 13: Parking queue statistics with Sunday deliveries allowed (95% CI, 4 replications).

The FC requires at least **2 empty-parking spots** and **5 full-parking spots**. The empty parking is virtually never occupied (average ≈ 0), meaning returned trailers reach a filling ramp almost immediately. The full parking averages 2.31 trailers, so roughly half its capacity is in use at any given time.

4.3. Average Time Each Trailer Spends at the FC

Ramp residence time was computed via a Visual Logic block that records the simulation clock at trailer entry into the connection activity and at exit from the disconnection activity. Queue times in both parkings were obtained from Simul8's built-in statistics. Table 14 summarises the results.

Stage	Minutes	Hours
Avg. time in Parking VIDE	0.00	0.00
Avg. time on filling ramps (connect + fill + disconnect)	217.58	3.63
Avg. time in Parking PLEIN	443.08	7.38
Total average time at the FC	660.66	11.01

Table 14: Average time spent by a trailer at the FC, with Sunday deliveries allowed.

The empty-parking wait is essentially zero: with only 10 trailers and three ramps, a free ramp is almost always available on arrival. The dominant component is the full parking (≈ 7.4 h), where trailers produced overnight accumulate until the first morning shift. Ramp time (≈ 3.6 h) is below the 5-hour theoretical maximum because many trailers return with residual hydrogen and require shorter fills. On average, a trailer spends roughly **11 hours** at the FC before dispatch.

4.4. Resource Utilisation Analysis

Table 15 presents utilisation rates for the optimal multi-tank configuration with Sunday deliveries allowed (2 empty, 3 full / 2 day, 3 night / 2,2,2,2,1 tanks), over 4 replications at 95% confidence.

Resource	Metric	Low 95%	Average	High 95%
Drivers (Chauffeurs)	Utilisation [%]	91.89	92.02	92.15
FC Operator	Utilisation [%]	5.75	5.77	5.78

Table 15: Resource utilisation rates for the optimal multi-tank configuration with Sunday deliveries allowed (95% CI, 4 replications).

Driver utilisation. At 92.02%, driver utilisation slightly exceeds the 90% upper threshold, placing the system at the edge of the fragility zone. In practice, the margin is narrow and the result confirms that driver capacity is fully exploited with virtually no idle time. This figure should be monitored closely in live operations; adding a single driver to the night shift should be considered if real-world variability proves higher than modelled.

FC Operator utilisation. At 5.77%, operator utilisation is very low, which is a direct consequence of modelling assumptions: connection and disconnection follow a Uniform(3, 8) min distribution (mean 5.5 min), while filling takes between 1.5 and 5 hours. With only three ramps, the operator spends most of their shift waiting. Utilisation could increase if the number of ramps were raised or if the actual connection procedure involves additional steps (pressure checks, safety verifications, paperwork) not captured in the current model. This result therefore reflects a deliberate modelling choice and should be revisited if more detailed process data become available.

4.5. Impact of Sunday Delivery Restrictions

4.5.1 Number of Empty and Full Trailers at the Start

When Sunday deliveries are banned, the system must pre-load all HRS during Saturday to cover Sunday’s full 20-hour consumption. All standard-scenario configurations fail under this restriction. The solution is to increase the on-site buffer at the HRS while keeping the same FC endowment and driver pool. Table 16 shows the result.

Trailers at FC		Drivers / shift		Tanks per HRS			Total trailers
Empty	Full	Day	Night	HRS 1–3	HRS 4	HRS 5	in system
2	3	2	3	3	2	1	17

Table 16: Optimal configuration under the Sunday delivery ban.

The three high-consumption stations (HRS 1–3, 1 000 kg/day) need a third tank to bridge the Sunday gap; HRS 4 (500 kg/day) stays at 2 tanks; HRS 5 (200 kg/day) stays at 1, since its low consumption gives each trailer over 70 hours of autonomy. The total fleet grows from 14 to 17 trailers; FC endowment and driver allocation remain unchanged.

The Sunday constraint cannot be resolved by adding more trailers at the FC or more drivers: with only two tanks at the high-consumption stations, the maximum on-site reserve (1 400 kg) is insufficient to cover a full Sunday plus the hours before the first Monday delivery reaches the most distant HRS. The bottleneck is storage capacity at the destination, not logistics, and only increasing HRS tank slots resolves it. The specific allocation

(3,3,3,2,1) was determined through systematic experimentation and is the minimum that guarantees zero stockouts under the ban.

4.5.2 Number of Parking Spots Needed

Table 17 shows the parking statistics under the Sunday ban (4 replications, 95% CI).

Parking	Metric	Low 95%	Average	High 95%
Parking VIDE	Average Queue Size	0.00	0.00	0.00
Parking VIDE	Maximum Queue Size	2.00	2.00	2.00
Parking PLEIN	Average Queue Size	2.52	2.53	2.53
Parking PLEIN	Maximum Queue Size	5.00	5.00	5.00

Table 17: Parking queue statistics under the Sunday delivery ban (95% CI, 4 replications).

Parking requirements are identical to the standard scenario (**2 empty spots, 5 full spots**). Average full-parking occupancy is marginally higher (2.53 vs. 2.31), reflecting Saturday pre-loading, but the peak is unchanged.

4.5.3 Average Time Each Trailer Spends at the FC

The same methodology was applied under the Sunday-ban scenario. Table 18 presents the results.

Stage	Minutes	Hours
Avg. time in Parking VIDE	0.74	0.01
Avg. time on filling ramps (connect + fill + disconnect)	215.96	3.60
Avg. time in Parking PLEIN	563.97	9.40
Total average time at the FC	780.67	13.01

Table 18: Average time spent by a trailer at the FC, under the Sunday delivery ban.

The total average rises from 11.01 to **13.01 hours** (+18%), almost entirely driven by the full parking (+27%): full trailers produced on Sunday accumulate undispached until Monday morning. Ramp time is virtually unchanged (215.96 vs. 217.58 min), as it depends only on physical filling parameters unaffected by the delivery ban. The small non-zero empty-parking time (0.74 min) reflects brief Monday-morning congestion when returning trailers occasionally saturate all three ramps simultaneously.

A. Simul8 Implementation Details

A.1. Data Structures

The model state is centralised in four Simul8 spreadsheets and three work item labels.

Spreadsheet `ss_trailers`. Each row corresponds to one HRS (rows 2–6 for HRS 1–5; row 1 holds headers). Columns store per-station parameters and real-time state:

Column	Content
1	HRS identifier (1–5)
2	One-way distance to FC [km]
3	Hourly consumption rate q_i [kg/h]
4	One-way travel time t_i [h] (computed at reset)
5	Current safe margin σ_i [h]
6	Number of trailer slots n_i at the HRS
7	Demand flag (1 = delivery en route, 0 = available)
8	Current aggregate hydrogen level L_i [kg]
9	Base travel cost $C_{\text{travel},i}$ (computed at reset)
10	Most recent cost-function value C_i

Table 19: Structure of `ss_trailers`. Rows 2–6 correspond to HRS 1–5.

Spreadsheet `ss_parameters`. Stores global constants and real-time flags. All values reside in column 2:

Row	Content
2	Trailer speed v [km/h]
3	Per-hour travel cost c_{travel}
4	Overtime penalty rate c_{overtime}
5	Rupture avoidance bonus B
6	HRS operating status (1 = on, 0 = off)
7	Night shift flag (1 = active, 0 = inactive)
8	Current simulation hour
9	Current day of the week (1 = Mon, ..., 7 = Sun)
10	Inefficiency cost c_{ineff}
11	Default safe margin σ_{default} [h]
12	Surplus cost c_{surplus}

Table 20: Key cells of `ss_parameters` (column 2).

Spreadsheet `ss_log`. Records a time series of hydrogen levels at every HRS for warm-up analysis and critical-period identification. One row is written per evaluation cycle from the Time Check Logic (Section A.3). A global variable `v_log_row` tracks the next row to write; it is initialised to 1 in the Reset Logic and incremented before each write, so data begins at row 2 (row 1 is reserved for headers).

Column	Content
1	Simulation time at the moment of recording [min]
2	Hydrogen level at HRS 1 [kg]
3	Hydrogen level at HRS 2 [kg]
4	Hydrogen level at HRS 3 [kg]
5	Hydrogen level at HRS 4 [kg]
6	Hydrogen level at HRS 5 [kg]

Table 21: Structure of `ss_log`. Each row is one snapshot per evaluation cycle.

Spreadsheet `ss_time_FC`. Records the time each trailer spends on the filling line (connection + filling + disconnection). A global variable `v_time_row` tracks the next row; it is initialised to 1 and incremented before each write, so data begins at row 2 (row 1 is reserved for a header).

Column	Content
1	Time the trailer spent on the filling line [min]

Table 22: Structure of `ss_time_FC`. Each row corresponds to one trailer that completed the FC cycle.

Work item labels. Each trailer carries three labels:

- **Destination:** the HRS identifier the trailer is assigned to (1–5), or 0 if unassigned.
- **Niveau:** the residual hydrogen level [kg] after a swap, or the flag 1 for a full 700 kg load in the full parking. Determines filling time at the FC via $t_{\text{fill}} = 300 \cdot (700 - \text{Niveau})/700$ minutes.
- **FC_Start:** timestamp (Simulation Time in minutes) recorded when the trailer enters the first filling activity (Connexion Rampe). Used to compute the filling-line residence time at exit.

A.2. Initialisation (Reset Logic)

The Reset Logic executes once at simulation time zero and establishes the starting state of the entire system. It declares four local variables: `index`, `Local_v_num_tanks`, `v_time_travel`, and `Local_v_cost_travel`.

Log and timing spreadsheet clearing. Both logging spreadsheets are reset. `v_log_row` and `v_time_row` are set to 1 so that the first write (which increments before writing) targets row 2, preserving row 1 for headers:

```

SET v_log_row = 1
SET index = 2
WHILE index <= 100000
    SET ss_log[1,index] = 0
    SET ss_log[2,index] = 0
    SET ss_log[3,index] = 0
    SET ss_log[4,index] = 0
    SET ss_log[5,index] = 0
    SET ss_log[6,index] = 0
    SET index = index+1

```

Global parameter loading. The trailer speed and per-hour cost are read from `ss_parameters`:

```

SET v_speed = ss_parameters[2,2]
SET v_cost_per_hour = ss_parameters[2,3]

```

HRS parameter initialisation. For each HRS (rows 2–6), derived quantities are computed from the base parameters stored in the spreadsheet:

```

SET index = 2
WHILE index <= 6
    SET Local_v_num_tanks = ss_trailers[6,index]
    SET v_distance = ss_trailers[2,index]
    SET v_time_travel = v_distance/v_speed
    SET v_cost_travel = v_time_travel*v_cost_per_hour
    SET ss_trailers[4,index] = v_time_travel
    SET ss_trailers[7,index] = 0
    SET ss_trailers[8,index] = Local_v_num_tanks*700
    SET ss_trailers[9,index] = v_cost_travel
    SET ss_trailers[10,index] = 999999
    SET index = index+1

```

For each station: the one-way travel time is computed as distance divided by speed (column 4); the demand flag is cleared (column 7); the initial tank level is set to $n_i \times 700$ kg, i.e. all slots start full (column 8); the base travel cost is stored (column 9); and the cost cell is initialised to 999 999 (column 10).

Parking initialisation. The two FC parkings are populated with their respective initial trailer counts (`C_vide` and `C_plein`, the decision variables N and M):

```

SET Parking VIDE.Initial Contents = C_vide
SET index = 1
WHILE index <= C_vide
    Select Current Work Item Parking VIDE , index
    SET Destination = 0
    SET Niveau = 0
    SET index = index+1

SET Parking PLEIN.Initial Contents = C_plein
SET index = 1
WHILE index <= C_plein
    Select Current Work Item Parking PLEIN , index
    SET Destination = 0
    SET Niveau = 1
    SET index = index+1

```

Empty-parking trailers receive Niveau = 0 (completely empty); full-parking trailers receive Niveau = 1 (full load flag). Both sets start with Destination = 0 (unassigned).

Timing spreadsheet clearing. Finally, `ss_time_FC` is cleared starting from row 2 (row 1 is the header):

```

SET v_time_row = 1
SET index = 2
WHILE index <= 150000
    SET ss_time_FC[1,index] = 0
    SET index = index+1

```

A.3. Time Check Logic

This block implements the greedy dispatching heuristic described in Section 1.3. It is triggered periodically throughout the simulation. Local variables declared: `Index`, `v_tank_level`, `v_consumption`, `Local_v_n_tanks`, `v_capacity`, `v_free`, `v_time_travel`, `v_demand`, `v_safe_margin`, `v_delivered`, `v_arrival`, `v_cost_shift`, `v_final_hour`, `Local14`, `v_hrs_min`.

Phase 1: System status. The block queries the two shift schedules, records the current time, and loads all global cost parameters into local variables:

```

Get Shift Status HRS , HRS_status
SET ss_parameters[2,6] = HRS_status
Get Shift Status Night Shift , Shift
SET ss_parameters[2,7] = Shift

SET v_current_hour = HOUR[Simulation Time]
SET v_day = DAY[Simulation Time]
SET v_overtime_cost = ss_parameters[2,4]
SET v_priority_bonus = ss_parameters[2,5]
SET v_penalty_per_kg = ss_parameters[2,10]
SET v_margin = ss_parameters[2,11]
SET v_cost_surplus = ss_parameters[2,12]
SET ss_parameters[2,8] = v_current_hour
SET ss_parameters[2,9] = v_day

```

Get Shift Status returns 1 if the named schedule is currently active and 0 otherwise. HRS covers the 04:00–00:00 operating window; Night Shift covers 13:00–21:00. If HRS_status = 0 (outside the operating window), the block exits immediately.

Phase 2: Station loop and safe margin. When the HRS shift is active, the block loops over all five stations. The minimum-cost tracker is initialised before the loop:

```

IF HRS_status = 1
  SET Index = 2
  SET Min = 999999
  WHILE Index <= 6
    SET v_consumption = ss_trailers[3,Index]
    SET v_time_travel = ss_trailers[4,Index]
    SET v_num_tanks = ss_trailers[6,Index]
    SET v_demand = ss_trailers[7,Index]
    SET v_tank_levels = ss_trailers[8,Index]
    SET v_cost_travel = ss_trailers[9,Index]

    IF Shift = 1
      IF v_day = 6
        SET v_safe_margin = [[3*v_time_travel]+4]+24
      ELSE
        SET v_safe_margin = 3*v_time_travel+4
    ELSE
      SET v_safe_margin = v_margin
  END WHILE
END IF

```

During the night shift (Shift = 1), the margin is $3t_i + 4$ hours; on Saturdays (v_day = 6), an extra 24 hours are added to pre-load for Sunday. Outside the night shift, the fixed default margin is used.

Phase 3: Tank level update. The hydrogen level at the current HRS is decremented unconditionally by its hourly consumption. If the level drops below zero, it is clamped and a stockout alert fires:

```

SET v_tank_levels = v_tank_levels-v_consumption
IF v_tank_levels < 0
    SET v_tank_levels = 0
    Display Message "HRS à sec"
SET ss_trailers[8,Index] = v_tank_levels
SET ss_trailers[5,Index] = v_safe_margin

```

Phase 4: Cost evaluation. The cost function is evaluated only for stations whose demand flag is zero (no delivery already en route). The projected arrival level and delivered quantity are computed first:

```

IF v_demand = 0
    SET v_capacity = v_num_tanks*700
    SET v_arrival = v_tank_levels-[v_time_travel*v_consumption]
    SET v_free = v_capacity-v_arrival
    IF v_free >= 700
        SET v_delivered = 700
    ELSE
        SET v_delivered = v_free

```

The cost is then assembled term by term. Travel cost plus inefficiency penalty:

```

SET v_cost_shift = v_cost_travel+[700-v_delivered]*v_penalty_per_kg
SET v_final_hour = v_current_hour+[v_time_travel*2]+0.5

```

Overtime component, applied relative to the relevant shift boundary:

```

IF Shift = 1
    IF v_final_hour > 21
        SET v_cost_shift = v_cost_shift+[v_final_hour-21]*v_overtime_cost
IF Shift = 0
    IF v_final_hour > 13
        SET v_cost_shift = v_cost_shift+[v_final_hour-13]*v_overtime_cost

```

Rupture component. If the projected arrival level falls at or below the safe margin (in kg), the large bonus B is subtracted, making the cost strongly negative and guaranteeing selection. Otherwise, the surplus autonomy beyond the margin is penalised:

```

IF v_arrival <= v_safe_margin*v_consumption
    SET v_cost_shift = v_cost_shift-v_priority_bonus
ELSE
    SET v_autonomy = v_arrival/v_consumption
    SET v_cost_shift = v_cost_shift+[v_autonomy-v_safe_margin]*v_cost_surplus

```

Phase 5: Candidate tracking and logging. The best candidate is tracked across iterations. The computed cost is also written to column 10 for monitoring:

```

        IF v_cost_shift < Min
            SET Min = v_cost_shift
            SET v_hrs_min = ss_trailers[1,Index]
            SET ss_trailers[10,Index] = v_cost_shift

    SET Index = Index+1

```

After the loop completes, the hydrogen levels at all five HRS are recorded in `ss_log`:

```

SET v_log_row = v_log_row+1
SET ss_log[1,v_log_row] = Simulation Time
SET ss_log[2,v_log_row] = ss_trailers[8,2]
SET ss_log[3,v_log_row] = ss_trailers[8,3]
SET ss_log[4,v_log_row] = ss_trailers[8,4]
SET ss_log[5,v_log_row] = ss_trailers[8,5]
SET ss_log[6,v_log_row] = ss_trailers[8,6]

```

This snapshot captures levels *after* the consumption decrement but *before* any dispatch, giving an accurate picture of the system state at each decision point.

Phase 6: Dispatch decision. Four conditions must all hold for a trailer to be dispatched: a valid candidate exists, the minimum cost is below 5 000, at least one full trailer is available, and at least one driver is free:

```

IF v_hrs_min > 0
    IF Min < 5000
        IF Parking.PLEIN.Count Contents > 0
            IF Chauffeurs.Current Available > 0
                Add Work To Queue Main Work Item Type , Queue Demand
                SET Destination = v_hrs_min
                SET ss_trailers[7,v_hrs_min+1] = 1

```

A new work item is injected into `Queue Demand` with `Destination` set to the selected HRS, and the demand flag for that station is set to 1. The +1 offset accounts for HRS identifiers starting at 1 while spreadsheet rows start at 2.

A.4. Swap Logic at the HRS (Work Complete Logic)

Each HRS has a Work Complete Logic block that executes when a trailer finishes the 30-minute swap activity. All five blocks share the same structure, differing only in the spreadsheet row index. The code for HRS 1 (row 2) is shown; HRS 2 through HRS 5 use rows 3 through 6 respectively.

```

SET v_num_tanks = ss_trailers[6,2]
SET v_tank_levels = ss_trailers[8,2]

IF v_tank_levels <= [v_num_tanks-1]*700
    SET Niveau = 0
    SET ss_trailers[8,2] = v_tank_levels+700
ELSE
    SET Niveau = v_tank_levels-[[v_num_tanks-1]*700]
    SET ss_trailers[8,2] = v_num_tanks*700

```

Two cases arise. **Full absorption:** if $L_i \leq (n_i - 1) \times 700$, the station can absorb the entire 700 kg delivery. The departing trailer leaves empty (**Niveau** = 0) and the aggregate level increases by 700 kg. **Partial absorption:** if the level exceeds that threshold, the departing trailer retains the excess hydrogen (**Niveau** = $L_i - (n_i - 1) \times 700$) and the aggregate level is capped at $n_i \times 700$ kg.

The **Niveau** label on the departing trailer directly determines its filling time at the FC: a fully consumed trailer requires the full 300 minutes; a trailer with residual hydrogen requires proportionally less.

A.5. Return Trip and Demand Flag Reset

When a trailer completes its return trip from an HRS to the FC, the corresponding **Travel Time FC Work Complete Logic** block resets the demand flag for that station, allowing it to be considered again in future cost evaluations. The code for HRS 1 is:

```

SET ss_trailers[7,2] = 0

```

The remaining four blocks are identical, targeting rows 3 through 6 for HRS 2 through HRS 5. The trailer itself re-enters the empty trailer parking at the FC.

A.6. FC Residence Time Tracking

To measure the time each trailer spends on the filling line (connection + filling + disconnection), two sets of blocks work in tandem.

Entry timestamp (Connexion Rampe — Route In After Logic). Each of the three connection activities records the simulation clock when a trailer enters the filling line. All three blocks are identical:

```

SET FC_Start = Simulation Time

```

From this moment, the trailer carries its entry timestamp as it progresses through the three filling stages.

Exit recording (Déconnexion Rampe — Work Complete Logic). Each of the three disconnection activities computes the elapsed time and writes it to **ss_time_FC**. All three blocks are identical:

```
SET v_time_row = v_time_row+1  
SET ss_time_FC[1,v_time_row] = Simulation Time-FC_Start
```

The counter `v_time_row` is incremented first, so each trailer occupies a distinct row. Since `v_time_row` starts at 1, the first data entry is written to row 2, preserving row 1 for the column header. The difference `Simulation Time – FC_Start` yields the filling-line residence time in minutes directly.

What is measured. The recorded time spans from entry into the connection activity to exit from the disconnection activity, and therefore includes the operator’s connection time, the filling duration, and the operator’s disconnection time. It does *not* include waiting time in either the empty parking (before connection) or the full parking (after disconnection). Those queue times can be obtained from Simul8’s built-in queue statistics and added separately if a total FC cycle time is desired.